High School Students' Perceptions of AI Use: Word Cloud and Statistical Analysis

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Abstract
This pilot study examines high school students' perceptions of artificial intelligence (AI) tools, using a mixed-methods approach that combines descriptive statistics and word cloud visualizations. Seven 11th-grade students in Turkey were surveyed about their AI tool usage, interests, future outlooks, and concerns. All participants had used AI (100%), unanimously citing ChatGPT as their tool of choice. Most reported using AI occasionally (monthly) for academic purposes (e.g. homework and research), with a minority using it weekly or daily. Qualitative responses suggest students view AI as a valuable aid for personalized and efficient learning, yet they express cautious optimism: common themes include enthusiasm for individualized support and time-saving benefits, alongside concerns about creativity decline, job displacement, and loss of control. We present filtered word clouds for open-ended questions to illustrate key terms (in the original Turkish, interpreted in English) and simple charts to summarize usage frequency and prevalent concern themes. These preliminary findings offer insight into how youth perceive AI's role in education and society, informing ongoing discussions on integrating AI in education in a responsible, human-centered manner[1][2].

Introduction

Generative AI tools (such as chatbots capable of natural language dialogue) have rapidly entered mainstream use in education. By 2023, large numbers of students and teachers were not only aware of AI tools like ChatGPT but also optimistic about their potential benefits. For example, a U.S. survey found that **68% of high school students** (ages 12–18) believed ChatGPT could help them become better, faster learners[3]. Teachers largely agreed, with 73% observing that AI can help students learn more[3]. Such positive perceptions reflect the promise that AI might **personalize learning** and accelerate academic progress. Indeed, the OECD notes that *"Generative AI has tremendous potential for education. Its biggest promise is personalised learning… offering experiences tailored to individual learners"*, though it cautions that *"these technologies also raise ethical considerations"*[2]. International organizations like UNESCO have similarly urged a **human-centered and governed approach** to AI in schools. In 2023 UNESCO called on governments to implement regulations and teacher training for AI in education, emphasizing public oversight to prevent harm[1]. Such calls underscore a dual reality: **AI in education offers great opportunities (e.g. adaptive tutoring, creative content generation) but also poses risks** (e.g. bias, privacy issues, erosion of academic integrity).

Amid this global context, there is a need to understand how students themselves perceive AI's growing role in their learning. Prior studies have begun to document student attitudes and usage patterns. Early evidence suggests students appreciate AI's utility for tasks like getting feedback, practicing skills, or generating ideas, yet remain wary of over-reliance and possible negative impacts[2][4]. This article contributes to that emerging knowledge base by reporting on a pilot survey of high school students' views on AI. We focus on **how often and why** students use AI tools, **what they find intriguing** about AI, and **which hopes or concerns** dominate their thinking. In particular, we integrate qualitative insights through open-ended questions, visualized via word clouds, to capture sentiment and ideas in students' own words. By analyzing this pilot data, we aim to identify key themes and issues that can inform larger-scale research and guide educators and policymakers in supporting youth as AI becomes increasingly embedded in education.

Method

Participants and Procedure

We conducted a pilot survey of high school students in Turkey in March 2025. The sample comprised $n = 7$ students (all in 11th grade, ~16–17 years old) enrolled in the science–math track (Turkish *"Sayısal"* curriculum). Participants were 6 female and 1 male, recruited as a convenience sample from a single school. Participation was voluntary and responses were collected anonymously via an online questionnaire (in Turkish). Given the small, homogenous sample and ongoing data collection, this study is framed as an exploratory pilot to refine the survey instrument and identify preliminary trends.

Survey Instrument

The survey included both closed-ended and open-ended items (translated from Turkish for analysis). Key sections of the questionnaire were: (1) demographics (grade level, academic track, gender), (2) prior experience with AI tools (yes/no and which tools), (3) frequency of AI use, (4) purposes of using or wanting to use AI, (5) most interesting aspect of AI, (6) outlook on AI's future, (7) anticipated role of AI in the education system, (8) biggest concern about AI, (9) hypothetical AI tool students would like to develop, (10) general comments on AI, and (11) any additional thoughts. Items 2–6 were multiple-choice or checkbox-style, whereas items 7–10 were open-ended prompts allowing free-text responses.

The survey was administered in Turkish. For analysis, responses to open-ended questions were translated into English. To preserve the authenticity of student voices, selected quotes are presented with English translations (the original Turkish phrasing is not included here, per the scope of this English-language report). We used simple **descriptive statistics** (frequency counts and percentages) to summarize the closed-ended responses. For the qualitative data, we employed **word cloud visualization** as an exploratory tool to identify prominent terms and themes in the text. Prior to generating word clouds, we applied filtering to remove very common words and context-generic terms (stop words in Turkish, as well as words like *"yapay"*/*"zeka"* meaning *"artificial intelligence"* and *"eğitim"* meaning *"education"*, which appeared in many responses by virtue of the questions). This filtering ensures the word clouds highlight the meaningful content of responses rather than the repetitive context. The word clouds were produced from the Turkish text of answers to questions 7–10, and their salient terms are interpreted in the results below. Figure references (Figure 1, 2, etc.) denote the visualizations embedded in the Results section.

It should be noted that word clouds offer a quick visual summary of frequently used words but do not capture the full nuance of responses (for instance, whether a word was used in a positive or negative context). Therefore, in interpreting the qualitative findings, we supplement the word clouds with narrative explanation and example quotes. Given the pilot nature of the study, our analysis is primarily descriptive and thematic. No complex statistical inference is attempted due to the small sample size. Instead, the emphasis is on identifying patterns that could inform future research with larger, more diverse student populations.

Results

AI Use and Tools Familiarity

All seven students reported that they **have used an AI tool** before (100% answered "yes" to having prior experience). Moreover, when asked which AI tools they have used, every student in this pilot **specifically mentioned ChatGPT** as a tool they had tried (7/7, 100%). No other AI tools (such as image generators or coding assistants) were cited by name in this sample; ChatGPT was evidently the common reference point for AI. This uniformity likely reflects ChatGPT's high profile and accessibility at the time of surveying. Students' **frequency of use** varied, but the most common pattern was occasional use rather than daily reliance. **Five out of seven** students (~71%) said they use AI tools roughly *"once a month"*, while one student (~14%) used AI about *"once a* *week"* and one student (~14%) reported using AI *"every day"*. **Figure 1** illustrates this breakdown of self-reported AI usage frequency. The predominance of monthly use suggests that for most of these 11th-graders, AI is an intermittent aid (e.g. for projects or occasional questions), whereas regular daily use was rare in this small sample.

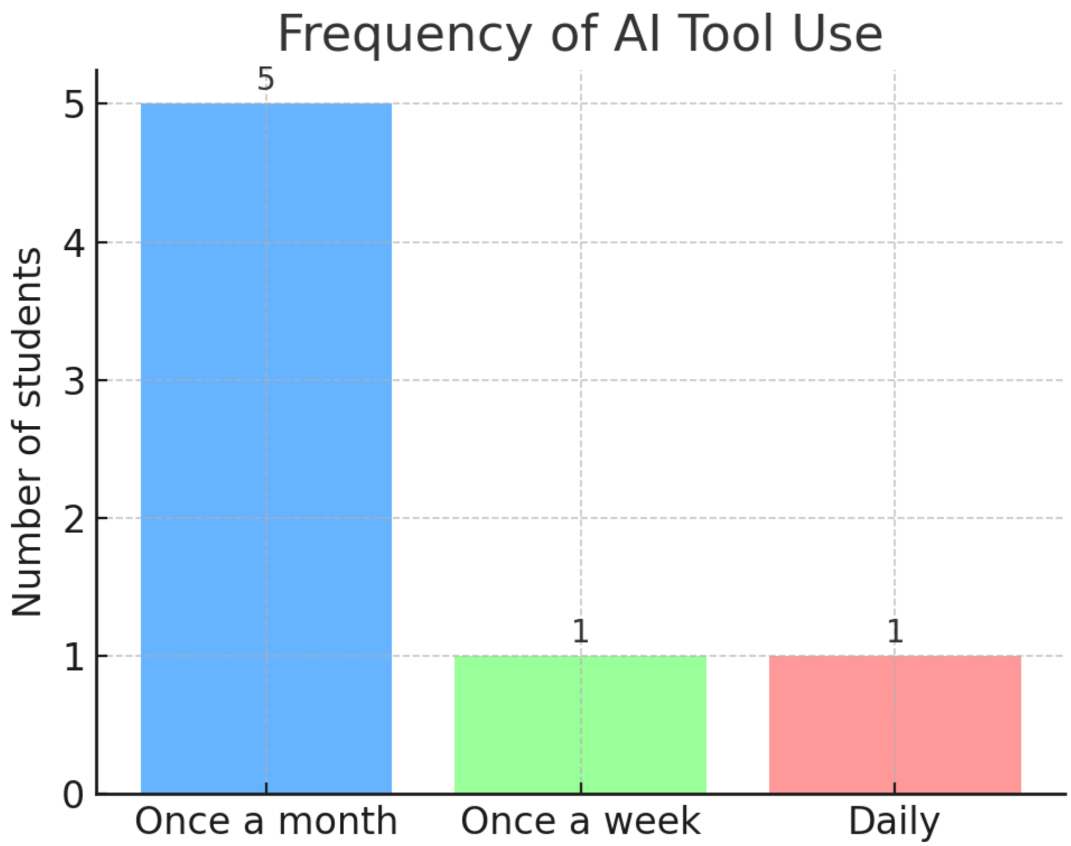

Figure 1. Self-reported frequency of AI tool use among the seven students. Most students use AI tools only occasionally (monthly), with only one daily user in this pilot group.

In terms of **purposes of use**, students indicated that they primarily turn to AI for academic tasks. A majority (5 of 7) said they use or would use AI tools for **schoolwork such as homework and research**. This was by far the most frequently mentioned purpose. A few students also identified other purposes: one student (14%) said they use AI for **creative writing** (e.g. composing stories or poems with AI assistance), and another (14%) mentioned using AI for **casual conversation or entertainment**. These responses suggest that, at least in this group, academic utility is the main driver of AI tool usage, with creative and entertainment uses being secondary and less common.

Students often described using ChatGPT to help with assignments – for example, to get explanations on a topic, brainstorm ideas for projects, or verify information for research. Notably, even those who used AI "for fun" still acknowledged academic support as a benefit. This aligns with broader observations that students view AI as a learning aid: surveys elsewhere have found that students believe tools like ChatGPT can supplement their study process and help them learn faster[3].

What Do Students Find Interesting About AI?

When asked about *"the aspect of AI that most interests you,"* students' answers varied, but two themes stood out. First, several students are intrigued by AI's ability to **support and enhance the learning process**. In particular, *three* of the seven respondents (43%) chose an option related to **AI helping their learning** (in Turkish: *"öğrenme sürecini desteklemesi"*, meaning *"supporting the learning process"*). This suggests that a significant portion of the students recognize and are interested in AI's potential to act as a tutor or study aid – for example, providing personalized explanations, practice problems, or feedback. The second common theme was **AI's future possibilities**: *two* students (29%) said they are most interested in *"the future possibilities and potential effects"* of AI. This indicates a forward-looking curiosity – these students are interested in how AI might evolve and impact the world (and their own futures) in years to come. Beyond these main themes, one student (14%) noted fascination with AI's capability to **generate visual media** (specifically mentioning image or video generation), and another (14%) was most interested in how AI can **converse like a human**.

In summary, the majority of students are drawn to AI either because of its educational utility or because of the broader innovative possibilities it represents. They see AI as *both a tool for personal learning enhancement and an emerging technology with transformative potential*. This dual interest reflects the common narrative around AI in education – it is valued for personalized learning opportunities while also captivating young people's imaginations about the future. As we will see in their comments, however, with this fascination also comes a measure of caution, especially regarding the future impact of AI.

Outlook on AI's Future

The survey also asked students, *"How do you see the future of AI technologies?"* Students' outlooks were mostly **optimistic with reservations**. A clear majority – *five* of the seven (71%) – characterized AI's future in positive terms *with the caveat that it must be used carefully*. The most common response (chosen by five students) was along the lines of *"AI's future is* positive*, but it should be used cautiously"*. This indicates that students anticipate AI will bring benefits and progress (such as advancements in education, science, daily life), yet they simultaneously acknowledge the need for *restraint or oversight* to avoid negative outcomes. Their written comments reinforce this balanced stance (e.g., *“a great development but there should be limits”* as one student wrote about AI – see general comments later).

One student (14%) was more **pessimistic or worried**, responding that they are *“concerned because there could be negative effects”* in the future. Another student (14%) had a distinctly negative view, stating *“(the future is) negative, because it threatens human jobs.”* This latter response explicitly mentions a fear that AI will displace human workers – a theme that will reappear in the discussion of concerns. Thus, while most of the class saw the glass as half-full (AI bringing improvements if handled responsibly), a minority already harbored fears of AI’s disruptive potential. Notably, none of the students described the future of AI in purely utopian terms (i.e. uncritically positive with no concerns); even the optimistic majority tempered their optimism with calls for careful use. This cautious optimism aligns with the perspectives expressed by educators and experts globally, who laud AI’s benefits for personalized learning but also highlight ethical pitfalls and the need for guidelines[2]. It is encouraging that students themselves echo this balanced view, recognizing both the promise and the perils of AI as they look to the future.

Open-Ended Responses: AI in Education, Concerns, and Personal Reflections

In addition to the structured questions above, the survey included four open-ended prompts (Q7–Q10) to elicit students’ own words about AI in education and their lives. These prompts asked about: Q7) AI’s role in the education system, Q8) their biggest concern about AI, Q9) what kind of AI tool they would create if they had the chance, and Q10) any general opinions about AI. Thematic analysis of these responses reveals insightful details about student sentiment. We present each question’s findings with supporting visualizations. The **word clouds** in Figures 2–5 highlight the most frequently used meaningful words in the Turkish responses (common function words and the phrase “artificial intelligence” itself have been filtered out). In interpreting these, larger words indicate ideas that multiple students mentioned or emphasized. We also summarize each theme with narrative and illustrative quotes (translated to English).

AI’s Role in the Education System (Q7)

When asked *“In your opinion, what role will AI play in the education system?”*, students predominantly envisaged AI as a tool for **enhancing and personalizing learning**. The majority of responses suggested that AI will make education more **individualized, interactive, and efficient**. For example, several students wrote that AI will *“make* *education more personalized and interactive"* for learners. Others echoed that AI could *"accelerate and improve the learning process"* for students. One student envisioned AI *"assisting teachers"* by handling certain tasks, *"but not replacing them."* These views reflect an expectation that AI will function as a *support system in classrooms* – tailoring educational content to student needs, providing instant help or enrichment, and generally augmenting the capabilities of teachers and learners.

However, not all students saw the impact as entirely positive. One student cautioned that AI *"might harm education because students could become lazy."* This perspective raises the concern that if AI makes things too easy or readily available, students might rely on it instead of learning the underlying skills (a worry about *over-reliance leading to complacency*). Nonetheless, this was a minority view in our sample; most students anticipated net positive contributions of AI in schooling, provided it is used wisely.

*Figure 2. Word cloud of responses to "What role will AI play in education?" (Q7). Common words like "education" and "AI" have been removed to highlight key terms. Students frequently mentioned concepts like "personalized" (*bireyselleştirilmiş*) and "interactive", as well as "accelerating" and "improving" the learning process. This*

suggests students expect AI to personalize learning and make it more engaging. A smaller segment warned of potential "harm" (e.g. making students "lazy"), indicating caution among some respondents.

Overall, the Q7 responses indicate that these high schoolers see a significant **supportive role for AI in education** – augmenting personalization, providing interactive learning experiences, and helping both students and teachers. They also implicitly assume that AI will be integrated into educational settings (classrooms or study routines) in the near future. The emphasis on personalization aligns with one of the central promises of AI in education noted by experts (that AI can adapt to individual learners' needs)[2]. At the same time, the mention of "lazy" students and not replacing teachers shows students are aware of the pitfalls (loss of student effort and the irreplaceable value of human teachers) and believe AI's role should be one of *assistance* rather than full automation of teaching.

Biggest Concerns About AI (Q8)

Students' responses to *"What is your biggest concern about AI?"* (Q8) clustered around **three major themes**. Despite the small sample, the same three concerns were repeatedly mentioned and, tellingly, they mirror common global debates about AI's societal impact. The primary concerns were:

1. **AI causing unemployment or job displacement** – 2 out of 7 students explicitly stated their biggest worry is *"AI creating unemployment"*. They fear that as AI automates tasks, human workers (including potentially themselves in the future) could lose jobs. One student had earlier noted the threat to human labor in their outlook on AI's future, reflecting this concern.
2. **AI reducing human creativity** – Another frequently cited concern (mentioned by 3 students) was *"that AI will diminish people's creativity"*. Students worry that over-reliance on AI for ideas or content might make people, especially their generation, less creative or intellectually lazy. For instance, if an AI can generate art, essays, or solutions instantly, individuals might not exercise their own imagination or critical thinking as much.
3. **AI getting out of control** – Two students raised the concern of AI *"going out of control"* or *"losing control"*. This encompasses fears of AI behaving unpredictably or even harmfully, beyond the limits set by its creators – a theme often seen in discussions about AI safety. They might be thinking of scenarios where AI systems make dangerous decisions, evolve beyond human oversight, or are misused, thus posing a threat.

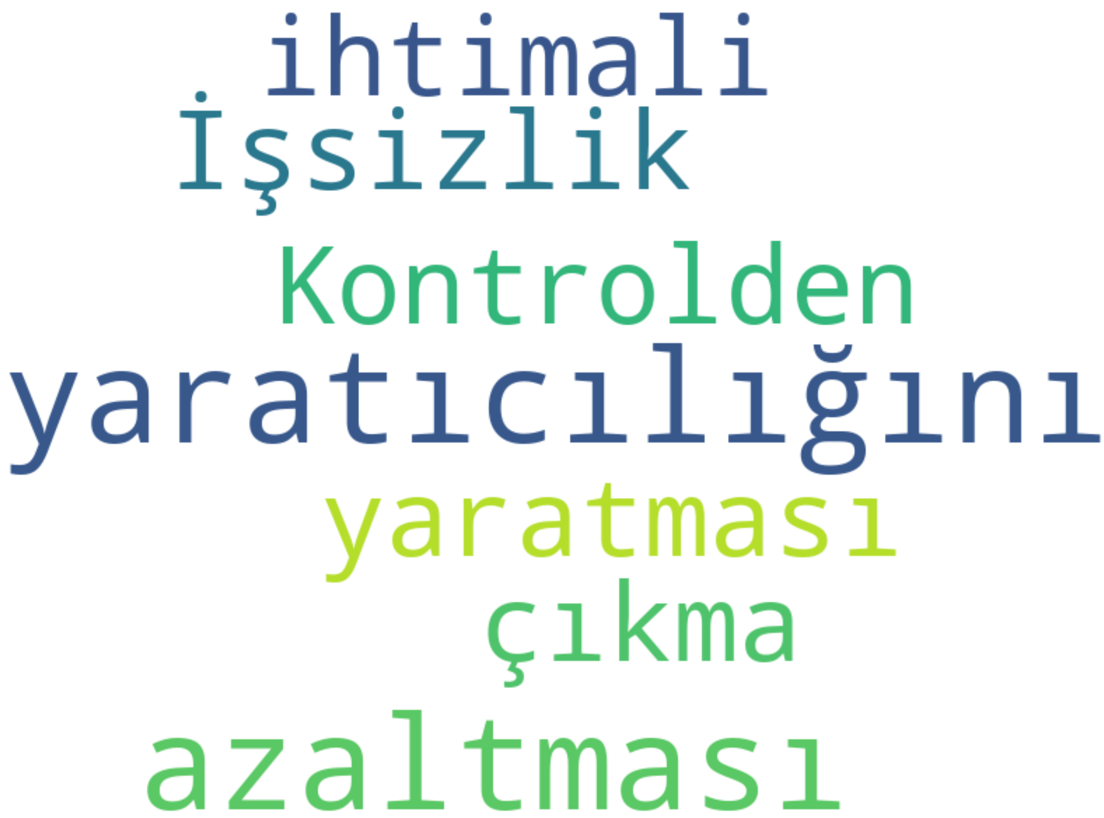

*Figure 3. Word cloud of responses to "What is your biggest concern about AI?" (Q8). Key words include "creativity" (*yaratıcılığını*, referring to people's creativity) and "reduction" (*azaltması*), highlighting the concern that AI may diminish human creativity. "Unemployment" (*işsizlik*) is also prominent, indicating fears of job loss. Words related to loss of control (e.g. "out of control" and "possibility" of it) appear as well. These three themes – creativity, jobs, and control – represent the dominant anxieties among the students.*

In quantitative terms, *loss of creativity* was the single most common worry (expressed by 3 students), followed by *job loss* (2 students) and *loss of control* (2 students). **Figure 4** shows the distribution of these concern themes by number of students mentioning each. Notably, each student named only one "biggest concern," so each respondent falls into one of these categories in this case. The fact that creativity loss topped the list is interesting – it suggests that students value human creativity highly and perceive AI as a potential threat to that human domain (perhaps they have in mind AI-generated art, writing, or even the ease of getting answers from AI which might discourage original thought). Concern about unemployment aligns with widespread discussions on AI and the future of work, indicating students are already aware of economic implications. And the fear of AI systems spiraling out of control reflects a more existential or safety-focused anxiety, which, while less common than the other two, shows up even in youth discourse (likely influenced by popular media or real news about AI incidents).

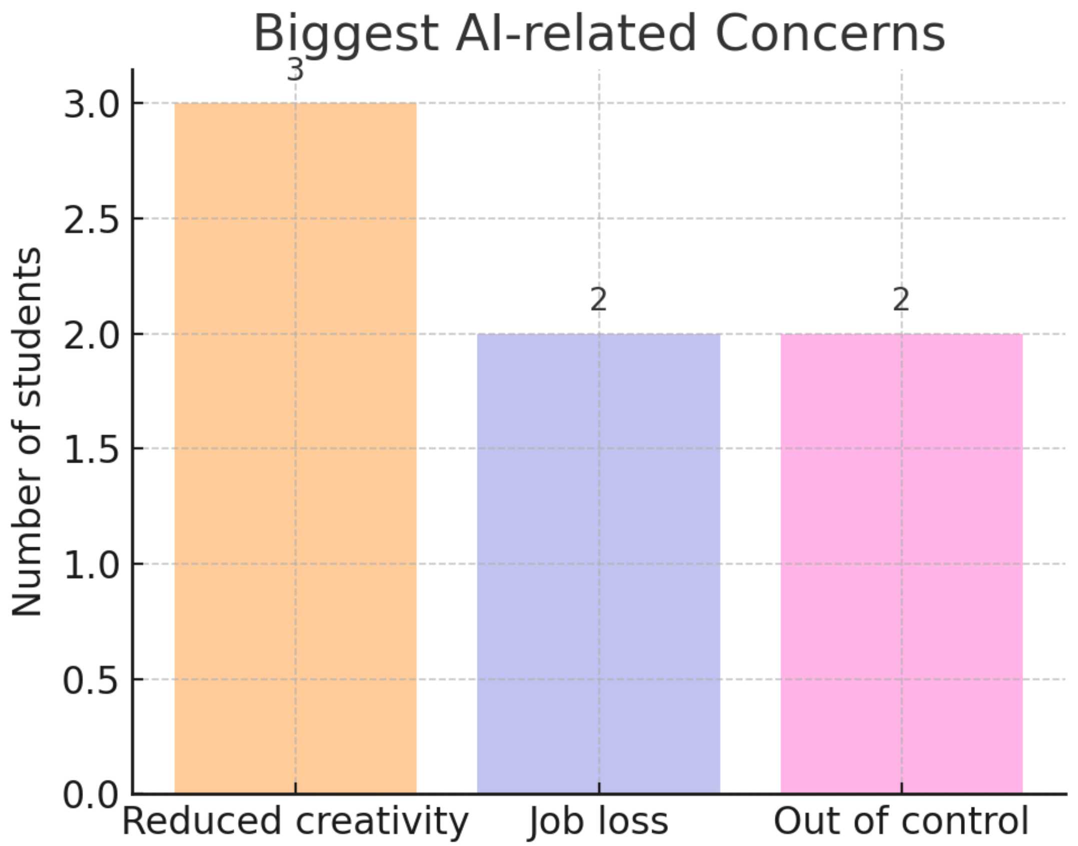

Figure 4. Main themes of students' stated AI concerns (Q8), by number of students who expressed each concern. "Reduced creativity" was the most frequently mentioned worry (3 students), followed by fears of "job loss" due to AI (2 students) and AI "going out of control" (2 students). These concerns align with broader societal debates on AI's impact on creativity, employment, and safety.

To illustrate, one student wrote their biggest concern was simply *"that it reduces people's creativity."* Another wrote, *"the possibility that [AI] could get out of control."* These succinct statements encapsulate the core of each fear. Importantly, such concerns are echoed by experts and institutions calling for AI ethics and governance; for example, persistent biases and unintended outcomes in AI systems can indeed erode trust and pose risks, underscoring why frameworks for accountability are needed[4]. Our students, in their own way, are voicing the need for AI to be managed so that human creativity, jobs, and safety are safeguarded.

It is worth noting that no student brought up concerns related to **privacy** or **data security**, which are also prevalent issues in AI discussions. This may be due to the specific phrasing of their responses or the immediacy of the concerns that first came to their minds (jobs, creativity, control). It could also reflect that, as high schoolers, they are more focused on personal and future livelihood implications than on abstract data privacy issues. Additionally, none explicitly mentioned **bias or fairness**, though the concern about AI running amok could tangentially include AI making unjust decisions. These absences highlight areas that educators might need to address, ensuring students are also aware of AI's implications for privacy and fairness. But within the scope of what they did express, it's clear that maintaining human **agency, creativity, and opportunities** in the face of AI advancement is a priority for them.

"If you could develop an AI tool, what would you want it to do?" (Q9)

This question invited students to think imaginatively about designing their own AI. The responses to Q9 were varied and often reflected each student's personal needs or interests, but a common thread was a desire for AI that provides **personalized assistance** in some domain. Several students envisioned AI tools that would help them in their **academic learning, tailored specifically to their individual needs**. For instance, one student wrote that they would want an AI tool *"to identify all my deficiencies in every course and create personalized video lessons or tests for me."* Another student similarly wished for an AI that could act as a personalized tutor, saying they'd like an AI that *"would, with a comprehensive assessment, serve as my personal dietitian and dermatologist"* – essentially a personalized health advisor combined with educational feedback (this response was unique, combining interest in healthcare and personal assessment). Yet another student simply said they'd like a **"personal assistant"** AI – an AI that could handle various tasks or help organize their life, essentially tailoring to their personal schedule and needs.

Figure 5. Word cloud of responses to Q9: "If you had the chance to develop an AI tool, what would you want it to do?" Prominent words include "özel" (meaning "special" or "personalized"*), "eksiklerimi tespit" (*"identify my deficiencies"*), "ders" (*"lesson"*), "videolar" (*"videos"*), and "testler" (*"tests"*). These reflect a strong desire for personalized educational support – students imagine AI that finds their learning gaps and creates custom materials (lessons, videos, practice tests). Other terms in smaller print indicate more varied ideas: "kişisel asistan" (*personal assistant*), "insansı işlevler" (*human-like functions*), "diyetisyenlik" (*dietitian services*), and even "kötülüğü önlemesini" (*"prevent evil"*), showing the range of creative concepts from practical study aids to broader helpful AI functions.*

Academic and personal tutoring functions were the most common, but there were also unique ideas: one student said they would develop an AI to *"make more human-like functions"* – implying an AI that is more humanlike in its behavior or interaction (perhaps an AI that can behave or think more like a person). And one particularly striking response was a student who wrote that they'd want an AI tool that *"prevents evil."* This brief answer suggests the student imagines an AI that could actively stop bad actions or mitigate threats in the world – a very broad and idealistic vision (possibly influenced by portrayals of AI as a powerful agent for good). It is interesting that while this student wanted AI to prevent harm, others were concerned about AI *causing* harm; it encapsulates the dual-use nature of AI – powerful technology that could be directed for either positive or negative ends.

The emphasis on **"bana özel"** (meaning *"for me"* or *"personalized"*) in multiple responses is notable. It indicates that students really value AI's capacity to **personalize experiences** – whether it's personalized lesson plans targeting one's weaknesses, or personalized advice in health or academics. This directly ties back to what many cited as the most interesting aspect of AI (supporting the learning process) and to their expectations for AI in education (individualized learning). Even at their young age, they recognize that a one-size-fits-all approach in education can be challenging, and they dream of AI that adapts to *them* individually. This finding resonates strongly with the broader narrative in AI-in-education research that personalization is a key benefit of AI tools; our students clearly see that promise and crave it in very concrete terms (e.g. "an instant video lesson on any topic I want, with notes and practice questions just for me").

In summary, the "dream AI tools" our participants described were largely **augmentative** – tools to augment their learning, health, or personal productivity. They were not, for example, asking for AI to play games or do something frivolous; even the more imaginative responses had an undercurrent of utility (preventing evil is a societal utility; being more human-like could be about better interaction). This suggests that high school students view AI as something that, above all, should *help people in practical ways*. They are thinking about how AI can solve problems or fill gaps they experience. Educators and developers might take note: students are essentially calling for AI tutors, AI study partners, and personal assistants. Pilot programs or tools that address these desires (while aligning with curriculum and ethical guidelines) could be well-received by learners.

General Opinions on AI (Q10)

Finally, students were invited to share any general thoughts about AI. These responses provide a window into the students' overall sentiment – their mix of excitement, appreciation, and caution regarding AI. Broadly, **students see AI as a positive development** that is already proving useful, but they also believe **clear limits and ethical boundaries** are needed. Several students wrote in praise of AI's usefulness. For example, one student commented that AI (ChatGPT in particular) is *"more useful than Google – it gives the answer instantly without unnecessary prolonging and saves time. I often get information from it when I need to write an essay."* This highlights how at least one student has integrated AI into her academic workflow (researching for essays) and perceives it as a time-saving, efficient resource. Another student said using AI is *"enjoyable… a place I like to spend time, and at the same time I like using AI in my* *academic work. I haven't encountered any problems so far."* This reflects a generally positive user experience – AI as both a helpful study tool and something fun or engaging to interact with. It's noteworthy that this student explicitly states they have not had any bad experiences ("no issues yet"), implying their concerns about AI are still theoretical or future-oriented rather than based on personal negative incidents.

On the other hand, many students tempered their enthusiasm with **warnings or conditions**. One wrote, *"It's a great development but there should be a limit to its use; especially it should not block the path of artists or steal their work."* This remark reveals a concern for the rights of content creators (a nod to ongoing debates about AI-generated art and intellectual property – the student is effectively saying AI should not plagiarize or undermine human artists' livelihoods). The call for "a limit" to AI use echoes the majority sentiment from Q6 that AI must be used carefully. Another student simply wrote, *"It should be restricted,"* as their general opinion – suggesting they feel AI's proliferation needs to be curbed or regulated in some way, perhaps to prevent misuse.

Some students also expressed **mixed temporal perspectives**: they see benefits now but worry about the future. For instance, one student said (paraphrasing their long comment): *In today's world, AI definitely makes many things easier and speeds up learning. However, for the future, I am sure its harm to the natural balance will be very dangerous.* Here the student acknowledges the immediate positive impact (efficiency, accelerated learning – which aligns with others' views) but then voices a serious concern for the long-term *"natural balance"*. This phrase "natural balance" might refer broadly to societal or even ecological equilibrium – possibly the student fears AI could disrupt how society functions or even affect humanity's relationship with nature. It's a thoughtful concern that goes beyond personal impact to a holistic view of AI's place in the world.

Finally, one student summarized their concerns by listing the common threats: *"I am of course worried due to the oft-mentioned threat factors like causing unemployment and possibly getting out of control."* This comment ties back directly to the top concerns identified in Q8 (job loss and loss of control), showing that the student is well aware that these are widely discussed issues ("oft-mentioned") and shares those worries.

Figure 6. Word cloud of general opinion statements (Q10). Students frequently used positive terms like "güzel" ("great/good"*, referring to AI as a great development), "faydalı"* ("useful"*), and noted "zaman"* ("time"*, in contexts of saving time). At the same time, words like "sınır"* ("limit"*), "kısıtlanmalı"* ("should be restricted"*), and references to "sanatçıların"* ("artists'" *[rights]) and "doğal denge"* ("natural balance"*) appear, highlighting students' calls for moderation and ethical use. The presence of both "olumlu"* ("positive"*) and cautionary words in their narratives underscores their cautious optimism – they appreciate AI's benefits but insist on responsible implementation.*

In summary, the general sentiments expressed by these students can be characterized as **cautious optimism**. They largely view AI as beneficial – making learning faster, providing enjoyment, and being a notable technological breakthrough ("a nice development"). Yet they are not blindly optimistic; nearly every student's comment included a "but" or a concern: *"but it shouldn't be overused," "but it must have limits," "but I worry about X."* This balance is precisely the kind of nuanced understanding that stakeholders hope to foster in AI literacy initiatives – recognizing advantages while remaining vigilant about risks. It is encouraging to see that high schoolers are already engaging with AI at this thoughtful level. They are essentially calling for **responsible AI**:

systems that augment human capabilities without undermining human creativity, livelihood, or ethical standards. This aligns with global recommendations that AI in education be **human-centred, with proper governance and risk management**[1][4]. Our students intuitively echo those principles, suggesting that with proper guidance, they can be key voices in shaping a future where AI is used constructively in their learning environments.

Discussion

Even with a very small and homogenous sample, this pilot study reveals several consistent themes in how high school students perceive AI. First, students **already embrace AI as a learning tool**. All participants had tried AI (ChatGPT) and most used it for school-related purposes. They see AI as a means to get help with homework, find information quickly, generate ideas, and generally *learn more efficiently*. This reflects the rapid penetration of generative AI tools into student life, and resonates with larger surveys where majorities of students report positive educational use of AI[3]. The fact that *personalization* came up repeatedly – in what interests them about AI, in how they envision AI's role in education, and in the AI tools they wish for – underscores that students value individualized support. They are essentially telling us that traditional one-size-fits-all education doesn't always meet their needs, and they recognize AI's potential to fill those gaps (e.g. by tailoring practice problems or explanations just for them). This finding aligns with the broader literature that heralds personalized learning as a key benefit of AI in education[2]. It also presents a clear directive: educational AI developers and educators should focus on *harnessing AI for personalization*, to deliver the kind of custom-tailored help that students want.

Second, students exhibit **cautious optimism** regarding AI's growing presence. They are not technophobic – on the contrary, they generally welcome AI (several explicitly praised its usefulness and said they enjoy using it). However, they are far from naive about its downsides. Nearly all students qualified their enthusiasm with calls for careful use, limits, or outright restrictions on AI in certain aspects. Their articulated concerns – that AI could make people lazy, undermine creativity, cause job loss, or behave dangerously – mirror the very concerns raised by ethicists and policymakers. For instance, the fear that AI could diminish human creativity and originality is a nuanced point for a teenager to raise, and it shows a keen understanding of how technology might impact human development. The worry about job displacement indicates that these students are already thinking about AI in the context of their future careers and societal structure. And the "out of control" scenario touches on AI safety issues that computer scientists grapple with (how to ensure AI systems remain aligned with human values and intentions). Such concerns validate the importance of ongoing efforts to set **ethical guidelines and governance for AI** in schools and beyond[1]. Students in our survey implicitly call for what UNESCO and others have formally recommended: clear policies on AI use (e.g. age-appropriate limits, academic integrity guidelines), and perhaps curriculum that addresses AI's implications so that young people are prepared to use it responsibly.

Another point of discussion is the **trust and transparency** issue. While not overtly mentioned by students, it underlies some of their concerns. For example, the fear of AI giving harmful outputs or making people lazy ties into whether students can trust AI to be a reliable aide and how much they should depend on it. Building *trustworthy AI systems* – those that are accurate, fair, and interpretable – is essential if these tools are to play a larger role in education[4]. Students are hinting that their trust in AI is conditional: they appreciate it when it's helpful and factual (like giving a quick answer correctly), but they remain wary of over-trusting it (hence the insistence that usage should be cautious). This is healthy skepticism. It suggests that an educational approach which encourages students to verify AI outputs, use AI as a support but not a crutch, and understand AI's limitations will likely resonate with them. Indeed, integrating AI literacy into education – teaching students how AI works, where it can go wrong (biases, hallucinations, etc.), and how to use it critically – would address many of the reservations they expressed.

The **alignment between student perceptions and expert perspectives** is a notable outcome of this pilot. Students' calls for maintaining human creativity and control echo the principles of human-centered AI ethics (ensuring AI augments rather than replaces human talents). Their interest in personalization echoes the investments being made in adaptive learning technologies. Their identification of potential negatives aligns with the need for risk management frameworks (such as the U.S. NIST AI Risk Management Framework) to mitigate issues like bias and unintended consequences[4]. This alignment is encouraging: it means that as the educational community formulates policies and tools for AI in schools, student viewpoints – at least from this small sample – are in harmony with many of the recommended directions (like *using AI to assist teachers and students, not replace them; leveraging AI for personalization; instituting guidelines to prevent misuse*). It will be important to continue involving student voices as stakeholders in these conversations, especially as we scale up to more diverse samples.

Limitations. This study is a pilot with significant limitations. Most obviously, the sample size is extremely small (only seven students) and not demographically diverse, which limits any generalization. All participants came from the same grade level and academic track in one country, and all had similar academic backgrounds. Therefore, the patterns observed here might not hold in a broader population – for instance, students in different countries or different socioeconomic contexts might have varying levels of AI exposure or distinct concerns. Additionally, participation was voluntary, which could bias the sample towards students who are more interested in technology or education (indeed, the largely positive attitude might reflect that those intrigued by AI were more likely to respond). The data rely on self-report, which carries honesty and interpretation risks (e.g., what one student calls “using AI once a month” might differ from another’s criteria).

The **word cloud method** itself has limitations in analysis. With such a small number of responses, a word appearing large in a cloud could simply be from one verbose answer rather than consensus. We mitigated this by filtering out trivial words and interpreting the clouds in combination with reading each response in full. Still, the word clouds should be viewed as illustrative visuals rather than rigorous content analysis. They do not capture negations or context – for example, the word “lazy” appears in Q7’s cloud, but it was mentioned in the context of a concern *against* AI overuse. Without context, one might misread that as students wanting laziness, which is not the case. We addressed this by explaining the context in text, but it underscores that qualitative coding (and perhaps counting themes manually) would be a more robust approach for a larger sample. In future work, performing a systematic thematic analysis or sentiment analysis on open-ended responses would yield deeper insights than simple word frequency.

Another limitation is the **translation** of responses. We translated Turkish answers to English for reporting, which risks losing some nuances (e.g., the formality or emphasis in phrasing). We attempted to preserve meaning accurately, but subtle connotations might not carry over. Presenting original quotes was avoided here as the target audience is English-speaking, but doing so in bilingual reports could improve authenticity.

Next Steps. This pilot is an early snapshot, and we are planning to expand the survey to a wider pool of students across different schools. A priority is to include schools from varied regions and socioeconomic backgrounds, to see if perceptions differ in under-resourced settings versus well-resourced ones. For example, do students with less access to technology view AI differently (perhaps placing more hope in it or conversely being more skeptical)? Does gender or academic specialization influence how AI is perceived (e.g., do science-track students differ from literature-track students in attitudes)? With a larger N, we can also examine potential correlations – such as whether heavier users of AI have fewer concerns, or if those who use it for creative tasks have different outlooks than those who use it strictly for homework.

Additionally, the survey instrument will be refined. The pilot revealed that some questions could be clarified – for instance, multiple-choice options could be expanded to capture more nuance (the “most interesting aspect” could allow multiple selections or include “none of the above”). We also noticed that no one mentioned certain issues like privacy; perhaps a direct question on data privacy or AI bias could be added to prompt thoughts on those matters. We plan to incorporate a simple **qualitative coding framework** alongside word clouds for the open-ended items, to systematically categorize responses into themes (as we narratively did with concerns, but in future with formal coding and intercoder reliability if possible). This will strengthen the analysis when scaling up.

Finally, it would be valuable to compare student perceptions with those of **teachers and parents**, as all three groups are stakeholders in the integration of AI in education. Some global surveys have done this comparison, finding interesting differences (for instance, teachers may be more optimistic about AI’s benefits than students in some cases, or vice versa)[5]. Understanding where students and educators align or diverge can help address gaps in expectations and guide professional development for teachers on AI tools. Given that all our pilot students identified ChatGPT as their sole used tool, it’s likely that teachers in their environment have not introduced a variety of AI applications in class yet. As schools develop policies (some are even banning AI tools, others encouraging them in moderation), knowing student sentiment will be crucial – students might resist outright bans if they see AI as helpful, or they might support restrictions if they themselves feel dependency could be harmful. Our pilot data suggest students would *welcome guidance* on AI use rather than blanket prohibition; they want to use AI, but responsibly.

Conclusion

In this exploratory study of seven high school students, we found that youth are already actively engaging with AI tools and forming nuanced opinions about their use. Students view AI as a **powerful aid for learning** – one that can personalize education and save time – and they are eager to leverage these advantages to become better learners. At the same time, they harbor prudent concerns about AI’s impact on human creativity, jobs, and the importance of maintaining control over technology. Rather than being swept up in hype or fear, these students exhibit a balanced outlook: AI is *“a great development”* in their eyes, **provided it is used thoughtfully and ethically**. They advocate for AI that empowers rather than replaces humans, echoing calls for human-centered AI integration in schools[1].

These preliminary findings, while not generalizable, offer a snapshot that can inform educators and policymakers. The fact that students themselves are calling for *caution and guidance* suggests that schools should neither ignore AI nor blindly embrace it, but rather implement AI use in a structured, well-supervised manner. For example, curriculum might include discussions about when using AI is appropriate versus when it undermines learning, echoing the very worries students raised about over-reliance. Schools could channel students' enthusiasm for personalized AI support by introducing vetted educational AI tools (for tutoring or practice) while also teaching about AI's limitations. Teacher training becomes crucial – as UNESCO recommends – so that teachers can help students navigate AI tools effectively and ethically[1].

In conclusion, high school students in this pilot are not passive subjects of the AI revolution in education; they are active thinkers and users formulating their own code of conduct for AI use. They largely welcome AI's assistance, envisioning it as a personal tutor or assistant, but they simultaneously urge moderation (in their words, *"there must be a limit"*). Future research with larger samples will further illuminate these perceptions, but even this small study highlights the importance of **listening to students' voices**. Their insights can guide the development of AI policies and tools that align with educational goals and ethical norms. By involving students in the conversation about AI in education, we increase the chances of achieving an outcome where AI truly becomes a beneficial partner in learning – one that *amplifies* human potential without diminishing it[2][4].

References

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